

Workshop 1: Databases

Entity-Relationship model and SQLite3

This workshop is dedicated to Databases with a focus on SQLite3. You will work in a group of 4-5 students. You need one computer per group (of course, if you want to do the coding in parallel you can use more than one). The idea is to share your code within the group. You can create a shared text file with all the code you will run during this session.

1) The conceptual model - duration: 20 minutes:

In this session, we will create an Entity-Relationship diagram to represent a computer database in a company. In the company, there are employees, we know their first name, last name, birthdate, phone, and email. There are also computers defined by a type ('laptop' or 'desktop'), a brand, a model, the RAM capacity, the disk capacity and the year they were bought. An employee can have multiple computers in the company but a computer is used by only one employee.

According to this small text, draw the "full/complete" Entity-Relationship Diagram to model the information.

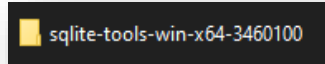
2) The logical model - duration: 10 minutes:

After having asked the teacher to look at your diagram and validate it, transform this conceptual model into a logical model (the list of tables in your database) according to the rules seen in lecture 04 – Databases.

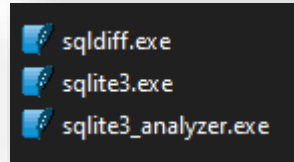
3) The database – part 1 – CLI - duration: 30 minutes

Please, download and install SQLite3 on your computer (<https://www.sqlite.org/download.html>). For Windows, the correct version is “sqlite-tools-...”. It is a zip (compressed file), **unzip it in a directory**. Ask for help if you don't know how to do it.

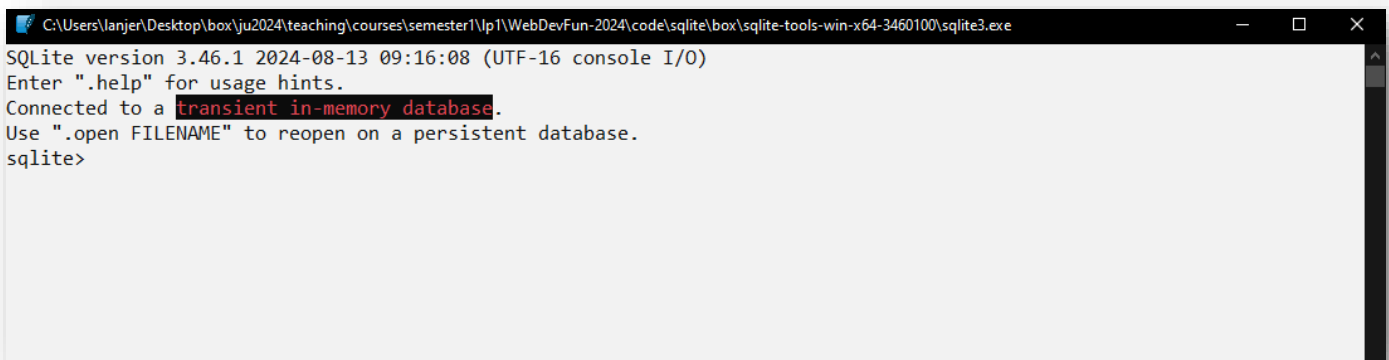
On Windows, you should obtain a directory named:



Enter this directory and run the file “sqlite3.exe”:



You should obtain a command line interface (CLI) window. In this CLI, you are connected to DataBase Management System (DBMS) SQLite3:



Congratulations/Grattis! You are ready to type in SQL commands after the “sqlite>” prompt.

Let's try something simple: SELECT 1+2;

```
sqlite> SELECT 1+2;
3
sqlite> _
```

If you get 3, it works because you asked SQL to display the result of a query (SELECT) and this query was a math expression: 1+2, so SQL did the job and computed 3. This “SELECT” is a SQL command, it needs to be terminated by “;” else it won't work.

Now that you know it works, you can try “.help” to see all the possible commands you can type.

```
sqlite> .help
.archive ...           Manage SQL archives
.auth ON|OFF           Show authorizer callbacks
.backup ?DB? FILE      Backup DB (default "main") to FILE
.bail on|off           Stop after hitting an error. Default OFF
.cd DIRECTORY          Change the working directory to DIRECTORY
.changes on|off        Show number of rows changed by SQL
.check GLOB            Fail if output since .testcase does not match
.clone NEWDB           Clone data into NEWDB from the existing database
.connection [close] [#] Open or close an auxiliary database connection
.crn1 on|off           Translate \n to \r\n. Default ON
.databases             List names and files of attached databases
.dbconfig ?op? ?val?   List or change sqlite3_db_config() options
.dbinfo ?DB?          Show status information about the database
.dump ?OBJECTS?        Render database content as SQL
.echo on|off           Turn command echo on or off
.eqp on|off|full|...   Enable or disable automatic EXPLAIN QUERY PLAN
.excel                Display the output of next command in spreadsheet
.exit ?CODE?           Exit this program with return-code CODE
.expert               EXPERIMENTAL. Suggest indexes for queries
```

The special commands in this CLI start with a dot (.help for instance) and you don't need a ";" at the end of these commands, they are not part of the SQL language but part of the CLI.

For now, you are working on an "in-memory database". It means if your computer crashes, nothing is saved on the disk and you will lose all your data. We will change that later on...

```
Connected to a transient in-memory database.
```

So, I recommend that you keep all the commands you will write into a text file somewhere on your computer to have the history of the commands (and then, you will be able to share them with the other students of your group). If you close SQLite3, you will lose your changes. That's why you need the SQL commands saved in a text file with your text editor (using Visual Studio Code is a good idea).

- a) Using the command line and the documentation (<https://www.sqlite.org/cli.html>), **create the table "Employee" (and only this one)** using this CLI.

```
CREATE TABLE Employee(...);
```

- b) When the table is created, insert a record (tuple, row) in it (**only one for now**):

```
INSERT INTO Employee VALUES (...);
```

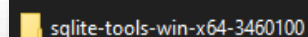
- c) Test if you can read and display the record:

```
SELECT * FROM Employee;
```

- d) At the end, you can save your work in a SQLite3 DB file:

```
.save company.sqlite3.db
```

You can check visually using the file explorer that your database has been created. By default, on Windows, it is located in the directory you extracted the tools:



```
sqlite-tools-win-x64-3460100
```

- e) When you know where this database file is, you can quit the CLI:

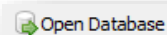
```
.quit
```

4) The database – part 2 – GUI - duration: 30 minutes

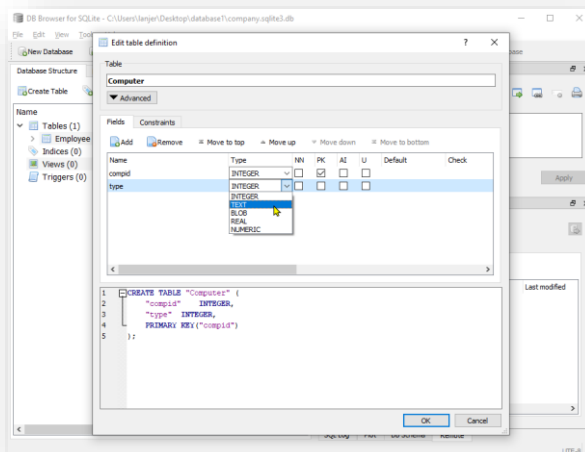
Download and install “DB Browser for SQLite” from <https://sqlitebrowser.org/dl/> on your computer. On Windows, please choose the “installer” version:

- DB Browser for SQLite – Standard installer for 64-bit Windows

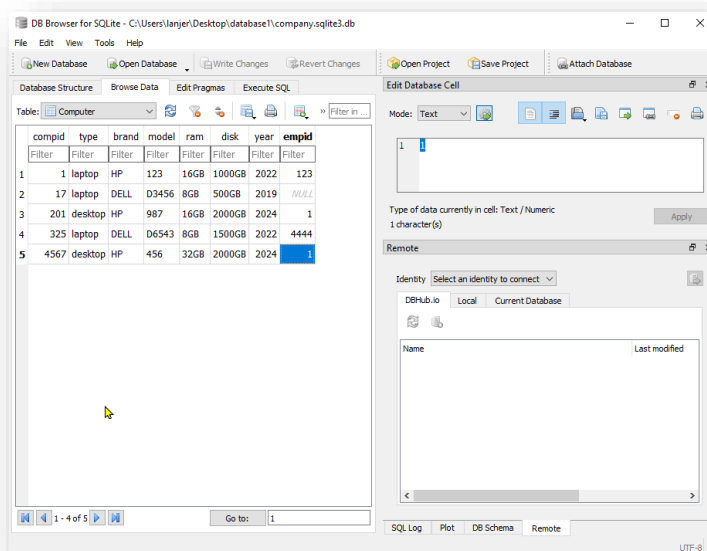
Using this tool, create the second table by opening the file “company.sqlite3.db”:



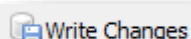
It should be easier with this Graphical User Interface (GUI). Create the table “Computer” using the buttons and forms:



With this GUI, add four employees and five computers to your database, assign the computers to the persons (randomly). You can have a computer not assigned to anybody with the “NULL” value set in the empid attribute (foreign key) in the “Computer” table.



Save your changes using:



When saved, you can close this CLI.

TACK OCH GRATTIS FÖR IDAG!